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Brazil and the Green Compute Arbitrage of the AI Decade

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Executive Thesis

Brazil is no longer best understood as a cyclical commodity exporter leveraged to global growth. That framing materially understates the structural shift now underway. Brazil is emerging as one of the most strategically advantaged geographies of the AI physical era, a country where energy abundance, mineral leverage, institutional depth, and a rare monetary setup converge to create a durable, long-horizon alpha opportunity.

As the United States and Europe increasingly collide with power-grid bottlenecks, multi-year interconnection queues, rising marginal electricity costs, and political friction around new infrastructure, Brazil is quietly moving in the opposite direction. Real rates are falling from restrictive levels, renewable generation capacity is scaling well beyond near-term domestic demand growth, critical minerals essential to AI hardware are being unlocked, and multi-gigawatt data-center campuses are being authorized at a scale that is increasingly unfeasible in OECD power markets.

The result is what can be described as Green Compute arbitrage: the conversion of surplus renewable energy and mineral endowment into globally tradable AI workloads. In this framework, Brazil is not merely a supplier to the AI buildout. It becomes a host economy for AI itself, capturing value not only from inputs, but from compute, infrastructure, and downstream digital activity.

What follows is grounded in physical constraints, capital structure, and macro dynamics, but also informed by firsthand experience living and working in Brazil, experience that shaped my conviction that the country's long-term strengths are often underestimated by outside investors.

Whenever I reflect on my career, there are many moments of luck along the path to who I am today. But no moment mattered more than moving to Brazil and running an office there less than five years out of college. I love the people, the country, the intensity, and the ambition that permeates Brazilian life. Those three years permanently altered how I think about emerging markets, institutions, and long-term potential.

I begin this paper there because it is rare to have the opportunity to write about a country you know personally at the precise moment when its structural advantages begin to align. I wrote a brief paper on Brazil in June 2025. Since then, the global physical AI upgrade cycle has accelerated materially. As compute, power, and capital constraints move from abstract concepts to binding limits, it is time to go deeper into why Brazil may be one of the most important and misunderstood beneficiaries of the AI era.

The Monetary Pivot: Real Rates, Inflation Psychology, and AI Deflation

Brazil has historically been “first in, first out” of global inflation cycles, and the current regime is no exception. After front-loading monetary credibility through aggressive tightening that drove policy rates to restrictive levels, Brazil now sits at the threshold of one of the most meaningful real-rate declines in the investable universe.

Consensus expectations point to policy rates falling materially over the next 12–24 months, while inflation expectations remain relatively well anchored. This places Brazil among a small and shrinking group of countries offering macro stability, positive real yields, and a clear easing trajectory.

This monetary setup matters disproportionately for AI and energy infrastructure. Data centers, transmission lines, rare-earth processing facilities, grid-scale storage, and power generation are long-duration, capital-intensive assets whose valuations are acutely sensitive to real discount rates. Even modest declines in real rates can materially lower the cost of capital, expand feasible project pipelines, and unlock equity re-rating potential across infrastructure-linked assets.

An underappreciated layer of this setup is the interaction between inflation psychology and the deflationary impulse of artificial intelligence.

I believe AI is the most powerful deflationary force the global economy has ever encountered. Outside of the physical commodities required for power generation, transmission, and compute infrastructure, AI is a deeply democratizing force, compressing costs, lowering barriers to entry, accelerating efficiency, and relentlessly reducing the marginal cost of production across services, software, logistics, and knowledge work.

This deflationary impulse stands in sharp contrast to the inflation psychology that still dominates policymaking and investor behavior in Brazil. The country's collective memory of hyperinflation remains deeply embedded in institutional decision-making, public discourse, and the risk premia demanded by investors. As a result, Brazilian rates continue to embed a meaningful inflation premium, one that reflects historical trauma as much as forward-looking fundamentals.

In an AI-accelerating world, this asymmetry matters. If AI-driven productivity gains suppress broad-based inflation over time while commodity-linked inflation remains localized to energy, metals, and infrastructure, the countries that have already priced inflation fear into their discount rates stand to benefit disproportionately. Brazil enters this transition with real yields that already compensate investors for inflation risks that may prove structurally overstated in an AI-driven economy.

This creates a powerful and unusual setup: declining real rates layered on top of an inflation premium that was built for a different era. For long-duration AI, energy, and infrastructure assets, this dynamic amplifies the convexity of Brazil's monetary pivot. The cost of capital falls not because inflation fears disappear, but because realized inflation increasingly diverges from the psychology embedded in rates.

In that sense, Brazil's inflation history, long viewed as a structural handicap, may quietly become a competitive advantage.

Breaking the Rare-Earth Monopoly: Brazil as a Strategic Mineral Hedge

Artificial intelligence is often framed as a software revolution. In reality, it is a mineral-intensive industrial transformation.

From GPUs and advanced cooling systems to humanoid robotics, precision motors, and automated logistics, the AI stack is fundamentally built on magnet metals, neodymium, praseodymium, dysprosium, and terbium. Today, these inputs remain overwhelmingly concentrated in China, creating a structural vulnerability in the global AI supply chain.

Brazil holds the world's second-largest rare-earth reserves and, after decades of underdevelopment, is now transitioning from geological potential to industrial relevance. Projects such as Serra Verde and Viridis Mining's Colossus deposit in Minas Gerais mark a shift from optionality to execution. Colossus, in particular, has demonstrated world-leading ionic recoveries of heavy magnet rare earths and cleared key regulatory milestones, positioning it as one of the most significant non-Chinese resources identified to date.

Execution risk remains, particularly around downstream processing, environmental oversight, and time-to-scale, but the strategic importance is clear. Every humanoid robot, industrial actuator, high-efficiency pump, and cooling system inside an AI data center is ultimately a magnet story.

Brazil's emergence as a non-Chinese supplier offers hyperscalers, OEMs, and sovereign buyers a geographically diversified hedge against supply-chain concentration risk. Equally important, Brazil is no longer content to export raw material. With several billion dollars of rare-earth-linked infrastructure now being unlocked, the country is moving downstream into alloys, magnet processing, and advanced materials capturing a greater share of AI's industrial surplus.

Power-Shoring AI: Brazil as a Global Green Battery

The most compelling pillar of the Brazil thesis is energy.

AI training is energy-inelastic. Large models do not wait for grid upgrades, zoning approvals, or political consensus. They go where power is abundant, reliable, and affordable.

Brazil operates an overwhelmingly renewable energy matrix, supported by hydro, wind, and a rapidly expanding solar base. This has created substantial surplus capacity on a curtailment-adjusted basis, capacity that, in many other jurisdictions, would remain stranded.

In contrast to the United States and Europe, where utilities face multi-year interconnection queues and growing opposition to new transmission, Brazil is authorizing multi-gigawatt grid connections to single AI campuses. The scale of announced projects reflects a deliberate strategy to monetize renewable surplus through digital infrastructure rather than allow it to be wasted.

These campuses function as programmable load centers. They absorb excess solar and wind generation that would otherwise be curtailed and convert it into globally tradable AI workloads. In effect, Brazil is turning renewable electrons into exportable intelligence.

Latency completes the picture. Sub-80-millisecond connectivity to North America allows Brazilian compute to serve global AI markets without meaningful performance degradation, strengthening the economic case for power-shoring AI training southward.

Creative Destruction and the New Commodity Supercycle

Brazil is uniquely positioned as the department store of the AI transition, a country supplying nearly every major physical input required for electrification, automation, and compute.

Iron and copper underpin global grid expansion, transmission, and data-center construction. Agriculture increasingly benefits from AI-driven precision farming, improving yields without expanding land use. Precious and strategic metals provide optionality in a world of currency debasement and geopolitical hedging.

This is not a traditional commodity supercycle driven by fixed-asset investment in China. It is a creative-destruction cycle, where AI simultaneously increases demand for minerals and improves the efficiency of discovering and producing them.

The same GPU clusters training foundation models in São Paulo and Rio are increasingly deployed to process satellite imagery, geophysical surveys, and exploration data across Brazil's underexplored shields. This creates a reflexive loop: AI accelerates mineral discovery, which supplies the materials needed to build more AI.

Few jurisdictions can replicate this dynamic at scale.

Talent, Institutions, and AI-Native Execution

Brazil brings more than rocks and rivers to the AI era. It brings institutional depth.

Home-grown champions such as Vale have trained generations of engineers, geologists, and operators in running high-throughput, low-grade operations across complex logistics environments. That expertise is now being redeployed into rare earths, copper, and AI-adjacent minerals.

Brazil's industrial workforce is already deeply AI-integrated. Predictive maintenance systems, autonomous fleets, and optimization algorithms have delivered meaningful efficiency gains. Surrounding the majors is a dense ecosystem of drillers, geophysicists, and technical service providers capable of moving from AI-generated targets to drilled resources on compressed timelines.

AI as a Human-Capital Multiplier

Beyond energy and minerals, AI is quietly transforming one of Brazil's most underappreciated assets: human capital. AI dramatically reduces the time, cost, and friction required to acquire complex skills, compressing the gap between countries with deep institutional pipelines and those with large, underutilized talent pools.

This matters disproportionately for Brazil. With a large, young, digitally fluent population and a strong culture of applied engineering across mining, energy, agribusiness, and industrial operations, Brazil is well positioned to benefit from AI-enabled education and training. Large language models, code copilots, simulation tools, and domain-specific AI assistants allow engineers, technicians, and operators to reach functional proficiency far faster than in prior cycles.

From an investment perspective, this represents a structural reduction in execution risk. The ability to train and upskill workers rapidly lowers operating costs, shortens project timelines, and expands the feasible scale of AI-adjacent infrastructure. As AI education becomes embedded in workflows rather than institutions, skill acquisition shifts from a front-loaded investment to a continuous process.

In effect, AI turns education into a deflationary input. The cost of knowledge falls, the speed of learning rises, and the return on human capital increases, reinforcing Brazil's appeal as a long-duration host for AI-intensive industrial activity.

The 2026 Election: Path-Dependent Risk, Not a Structural Break

Brazil enters the October 2026 general election with heightened investor attention. The election should be understood as a tempo and execution variable, not a referendum on Brazil's structural advantages.

Fiscal debates and regulatory uncertainty may affect sequencing and sentiment, but there is unusual cross-party alignment around monetizing renewable surplus and critical minerals through long-duration digital infrastructure. Data centers, grid investments, and strategic minerals are broadly viewed as mechanisms to attract foreign capital and anchor higher-value industrial activity domestically.

The election affects timing and discount rates not the validity of the Green Compute thesis itself.

Incentives and Capital Architecture: Stacking the IRR

Brazil offers a layered incentive stack tailored to Green Compute, including tax relief, accelerated depreciation, and long-tenor development-bank financing for grid-linked projects.

For hyperscalers and infrastructure investors, these incentives materially improve after-tax project IRRs relative to equivalent builds in power-constrained OECD markets, even before accounting for Brazil's lower levelized cost of energy.

Bottom Line

Brazil is no longer a simple beta play on global commodities. It is a structurally advantaged, AI-native economy positioned at the intersection of declining real rates, renewable abundance, mineral scarcity, and accelerating global demand for compute.

There is a long-standing joke that Brazil is the country of the future, and always will be. But in a world where AI is constrained not by algorithms, but by power, materials, and capital, the future has finally run out of alternative venues. Brazil offers all three at scale and at the right point in the cycle. The "future" is no longer a punchline; it's a physical necessity.

Having watched Brazil's ambition, ingenuity, and resilience up close early in my career, it is striking to see how those same qualities now align with the physical and human requirements of the AI age, turning long-held potential into a tangible inflection point.

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